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Correct Answer

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Practice Exam - 1

Question 104 of 117



A 54-year-old woman presents with an anterior ST-elevation myocardial infarction (MI) complicated by cardiogenic shock. She undergoes successful primary percutaneous coronary intervention. Post PCI, she remains hypotensive with BP 78/50 mmHg with SGC numbers demonstrating a PCWP of 34 mmHg and CI of 1.8L/min/m² despite initiation of three vasopressors (8 mcg/min of epinephrine, 22 mcg/min of norepinephrine, and 30 ng/kg/-min of Angiotensin II). Based on data from the Danger Shock trial, an Impella CP was placed. After 30 minutes of support, blood pressure was unchanged and she developed cool extremities, oliguria, and a rising lactate from 3.0 to 6.5 mmol/L. Bedside echocardiogram shows a left ventricular ejection fraction of 15% without mechanical complications of acute MI.

What is the most appropriate next step in management?

- A. Increase the norepinephrine infusion and reassess hemodynamics in 6 hours
-  B. Continue current inotrope/vasopressor therapy and continue to monitor lactate
-  C. Emergent initiation of Veno arterial extracorporeal membrane oxygenation (VA-ECMO) as a bridge to decision
- D. Proceed directly to durable left ventricular assist device implantation without temporary support
- E. Upgrade Impella device to 5.5 if LA doesn't improve over the next 12 hours

Commentary References

Content Code (Blueprint): Critical Care Cardiology — Cardiogenic Shock / Mechanical Circulatory Support (AHFTC Blueprint)

Task: Management

Teaching Point: Veno arterial extracorporeal membrane oxygenation (VA-ECMO) should be considered for refractory cardiogenic shock with worsening end-organ perfusion despite maximal medical therapy and impella CP support, as a bridge to recovery, decision, or durable therapy.

Correct Answer (Best Answer Choice): Emergent initiation of Veno arterial extracorporeal membrane oxygenation (VA-ECMO) as a bridge to decision

Distractors (Plausible Incorrect Options):

- Increase the norepinephrine infusion and reassess hemodynamics in 6 hours
- Continue current inotrope/vasopressor therapy and repeat lactate in 24 hours
- Proceed directly to durable left ventricular assist device implantation without temporary support
- Add a second intra-aortic balloon pump for additional afterload reduction

Rationale:

Correct Answer: C – “Emergent initiation of veno-arterial (VA) ECMO as a bridge to decision”. This patient has SCAI Stage D/E (deteriorating/extremis) cardiogenic shock with evidence of worsening end-organ hypoperfusion despite inotropes, vasopressors, and Impella CP support. VA-ECMO provides immediate full circulatory providing up to 6-7 L of support (although often run at lower flow rates) as well as gas-exchange support, allowing time for myocardial recovery or determination of candidacy for durable LVAD or transplantation.

Incorrect answers:

- A. Increasing norepinephrine dose: Escalating catecholamines in a patient already failing maximal pharmacologic support increases myocardial oxygen demand, arrhythmia risk, and end-organ vasoconstriction without addressing the underlying pump failure. Increased vasopressor use has been associated with worsening mortality.
- B. Continue current therapy and reassess in 24 hours: Delaying adequate mechanical circulatory support in a deteriorating shock state risks progression of detrimental pathophysiologic processes such as vasodilation or vasoconstriction, microcirculatory dysfunction inflammation with concomitant irreversible end-organ injury and death; SCAI staging emphasizes early escalation when trajectory is worsening.
- C. Correct answer
- D. Durable LVAD without temporary support: Implanting a durable device in a patient with a reasonable chance for myocardial recovery is not indicated and could cause harm rather than benefit. Further, even if myocardial recovery was deemed unlikely and durable LVAD was the overall long-term plan, LVAD implantation in an unstable, acidotic patient with unclear neurologic and end-organ status carries prohibitive perioperative risk; temporary support should bridge to a more informed decision.
- E. Delaying adequate circulatory support for 12 hours is not recommended as per answer B above. Further, the patient is failing 3 vasopressors and an Impella CP and so Impella 5.5 would likely be inadequate level of support.